

TULSA JONES, OK, USA

WMO: 723564

Lat: 36.039N

Lon: 95.984W

Elev: 638

StdP: 14.36

Time Zone: -6.00 (NAC)

Period: 97-19

WBAN: 53908

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	14.5	18.7	2.1	6.3	19.2	6.8	8.0	23.5	21.2	54.0	19.6	54.5	5.7	20	0.460

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.2	100.6	75.9	98.3	76.6	95.3	76.4	79.6	93.5	78.5	92.0	77.7	91.0	8.9	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.8	138.2	85.6	74.9	134.0	85.0	73.6	128.2	84.1	43.6	93.5	42.5	92.4	41.5	91.1	85.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.7	17.7	16.1	DB	5.7	104.1	5.4	3.9	1.8	106.9	-1.4	109.2	-4.4	111.4	-8.4	114.2	(4)
(5)			WB	4.6	81.5	5.0	1.9	0.9	82.9	-2.0	84.1	-4.9	85.2	-8.5	86.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	61.3	38.6	42.5	51.8	61.1	69.7	78.5	82.8	82.1	74.2	62.2	50.7	40.5
	DBStd	17.74	10.19	10.81	10.36	8.51	7.39	5.34	4.87	5.20	7.03	8.76	9.51	9.69
	HDD50	1180	376	254	101	12	0	0	0	0	0	10	106	321
	HDD65	3455	820	631	423	171	41	1	0	0	11	160	436	761
	CDD50	5310	21	45	157	345	611	854	1018	995	725	387	126	26
	CDD65	2108	0	2	14	54	186	405	553	530	285	72	6	1
	CDH74	24160	3	21	137	551	1811	4579	6949	6357	2953	721	76	2
	CDH80	11849	0	3	23	130	617	2163	3834	3496	1358	216	9	0
Wind	WSAvg	6.5	6.5	7.1	7.9	8.1	7.1	6.9	5.6	5.2	5.4	5.9	6.4	6.2
Precipitation	PrecAvg	39.6	1.5	1.7	3.1	4.3	5.4	5.0	3.5	3.1	4.1	3.2	2.6	2.3
	PrecMax	57.4	3.8	4.0	7.7	9.8	13.1	10.2	9.2	10.1	9.7	8.3	7.2	8.1
	PrecMin	27.0	0.1	0.1	0.8	1.5	0.9	1.1	0.3	0.0	1.1	1.1	0.0	0.1
	PrecStd	7.8	1.0	1.0	1.7	2.0	2.8	2.4	2.2	2.2	2.1	1.9	2.0	1.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	72.7	77.0	82.9	88.3	92.6	99.4	104.4	106.3	101.6	90.3	81.1	72.3
		MCWB	55.7	59.2	63.6	66.8	73.5	76.0	74.9	75.3	74.4	71.9	64.2	61.5
	2%	DB	66.4	71.6	77.8	83.6	89.7	95.7	100.5	101.1	94.7	85.8	75.6	66.5
		MCWB	53.4	57.9	61.2	66.3	72.4	76.7	75.9	76.2	74.5	69.5	61.0	56.0
	5%	DB	62.2	66.4	73.4	80.5	86.7	93.0	98.1	98.7	91.4	81.7	71.8	62.4
		MCWB	51.9	55.2	59.7	64.5	71.4	75.7	77.0	76.6	73.8	66.7	59.9	53.2
	10%	DB	56.0	62.0	69.9	76.7	83.5	90.6	95.1	95.3	88.2	77.8	67.2	57.0
		MCWB	46.7	51.9	58.2	63.3	70.4	74.8	76.8	76.4	72.5	65.3	57.3	49.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	61.8	64.4	67.1	71.5	77.1	80.4	81.3	81.0	78.0	74.3	67.6	64.4
		MCDB	67.1	72.0	78.6	82.4	88.2	93.3	94.0	95.1	91.0	85.4	75.8	70.2
	2%	WB	57.1	61.2	64.6	69.3	75.1	78.6	79.8	79.4	76.6	72.4	64.5	59.6
		MCDB	63.4	68.5	74.0	79.0	85.2	91.1	93.5	93.9	89.6	81.7	71.1	64.6
	5%	WB	52.0	57.3	62.3	67.7	73.5	77.5	78.6	78.2	75.3	70.5	62.0	54.7
		MCDB	60.3	64.9	71.0	76.2	83.3	89.3	92.2	92.5	87.8	78.0	69.3	59.2
	10%	WB	47.2	52.0	59.7	65.7	72.0	76.2	77.8	77.2	74.0	67.8	59.0	50.1
		MCDB	55.1	60.8	68.3	74.2	81.2	87.5	91.2	91.0	85.1	75.2	66.2	57.1
Mean Daily Temperature Range	5% DB	MDBR	21.9	22.1	22.5	22.6	20.6	19.8	21.2	22.3	22.6	23.1	23.2	20.9
		MCDBR	31.9	29.9	28.2	27.2	23.6	22.4	25.4	27.4	25.9	27.1	28.7	28.9
		MCWBR	20.3	18.8	15.6	13.8	9.9	7.8	7.4	7.9	8.9	12.8	16.7	19.0
	5% WB	MCDBR	27.0	25.8	22.2	21.7	19.6	19.8	21.7	22.7	21.8	20.7	23.7	22.5
		MCWBR	19.3	18.8	13.6	13.3	9.9	8.4	8.0	8.0	8.2	11.5	16.7	18.2
Clear-Sky Solar Irradiance	taub	0.302	0.309	0.352	0.387	0.425	0.433	0.442	0.443	0.395	0.352	0.318	0.298	
	taud	2.494	2.489	2.398	2.318	2.274	2.281	2.298	2.258	2.365	2.472	2.510	2.518	
	Ebn at Noon	285	297	290	284	274	270	267	264	272	277	277	280	
	Edh at Noon	26	29	35	40	43	43	41	42	36	30	25	24	
All-Sky Solar Radiation	RadAvg	818	1002	1320	1646	1813	2077	2095	1896	1582	1185	870	693	
	RadStd	82	150	80	105	172	203	121	125	133	149	107	83	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (38 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+173
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page